Toward a four-dimensional Yang–Mills construction

Notebook 10: local curvature perturbations at a fixed constraint

Research working document

16 October 2026

Abstract

We identify the orthogonal projection onto feasible curvature variations as two curl derivatives of the constrained flat inverse. The full inverse estimates of Notebook 09 prove that this projection is exponentially local in physical cell Hilbert norm, uniformly in refinement and compatible periodic volume. This gives a convergent weighted expansion for small local perturbations of the curvature metric, with explicit factorial and source-path bounds at every order. The averaging constraint is held fixed throughout. Identifying all coefficients of an actual nonflat Yang–Mills background remains a separate problem.

Status. Research proof checked by a separate internal derivation and audit. No external certification or novelty claim is made. The homogeneous flat input operators are those matched to the pinned source account in Notebooks 08 and 09. Fine spacing is M^{-1} for odd M, physical coarse spacing is one, and dimension is fixed with $d = 4$ the intended application. All estimates below use the same physical normalizations as those notebooks.

1 Local curvature perturbations at a fixed averaging constraint

This section concerns a flat linear problem with its averaging map held fixed. It supplies a full weighted analytic neighborhood for perturbing the curvature quadratic form. It does not identify a non-Abelian background variation, which can also change the averaging constraint, covariant derivatives, gauge projection, lower-order terms, and current.

Let D be the fine scalar gradient, d the fine curl, and $\mathcal{L} = D^*D$. Fix $Q = Q_M^{\mathrm{st}}$ and its scalar counterpart Q_0, so that $QD = d_c Q_0$. All fine Hilbert norms use physical fine volume, and the coarse cell norm uses counting measure. Retain the matched scalar source projection R and scalar inverse G', with

$$Q_0 G' R = 0, \qquad \mathcal{L} G' R = R. \tag{1}$$

For a fixed positive penalty a, let

$$\mathcal{A}_a = d^*d + DRD^* + Q^*aQ, \qquad G_0 = \mathcal{A}_a^{-1}, \qquad q_{\mathrm{aux}} = (QG_0Q^*)^{-1} = q_M + aI. \tag{2}$$

Here q_M is the physical Maxwell minimum and q_{aux} is the distinct penalized coarse precision. The matched unit-scale source has $a = 1$. Let

$$H = G_0 Q^* q_{\mathrm{aux}} = H_{\mathrm{src}}, \qquad F = dH. \tag{3}$$

Thus $QH = I$, $RD^*H = 0$, and $q_M = F^*F$. The full inverse theorem supplies uniform exponential cell L^2 locality for G_0 and all its fine derivative sandwiches of total order at most two. The minimizing extension theorem supplies the same cell locality for F. These inputs hold uniformly in odd M and on

every compatible periodic torus, as well as in homogeneous infinite volume. No assertion for general Dirichlet boundaries is used here.

For cell-operator kernels, use the weighted algebra norm

$$\|T\|_\kappa = \max\left\{\sup_x \sum_y e^{\kappa\,\mathrm{dist}(x,y)}\,\|T(x,y)\|\,,\quad \sup_y \sum_x e^{\kappa\,\mathrm{dist}(x,y)}\,\|T(x,y)\|\right\}. \tag{4}$$

The distance is ℓ^1 cell distance, or its periodic version. The internal norms in (4) are operator norms between the physical-volume cell Hilbert spaces; this convention also applies to maps between coarse bond cells and fine curvature cells. The norm is submultiplicative and unchanged by adjoints.

Theorem 1.1. *Let $\mathcal{K} = \overline{d\ker Q}$ in the curvature Hilbert space, and let $P_\mathcal{K}$ be its orthogonal projection. Then*

$$\Gamma = G_0 - H q_{\mathrm{aux}}^{-1} H^*, \qquad P_\mathcal{K} = d\Gamma d^* = dG_0 d^* - F q_{\mathrm{aux}}^{-1} F^*. \tag{5}$$

For a sufficiently small fixed $\kappa > 0$, the constants

$$f = \|F\|_\kappa\,, \qquad p = \max\left\{1, \|dG_0 d^*\|_\kappa + f^2\,\|QG_0Q^*\|_\kappa\right\} \tag{6}$$

are bounded independently of refinement and compatible periodic volume, and $\|P_\mathcal{K}\|_\kappa \le p$.

For any complex curvature operator V satisfying $p\,\|V\|_\kappa < 1$, the expression

$$q(V) = q_M + F^* V (I + P_\mathcal{K} V)^{-1} F \tag{7}$$

is a holomorphic weighted-cell-operator map. On the real selfadjoint slice, it equals the constrained positive-form minimum

$$\langle B, q(V) B\rangle = \inf_{QA=B} \langle dA, (I+V) dA\rangle. \tag{8}$$

It obeys

$$\|q(V) - q_M\|_\kappa \le \frac{f^2\,\|V\|_\kappa}{1 - p\,\|V\|_\kappa}, \tag{9}$$

$$\|q(V) - q_M - F^* V F\|_\kappa \le \frac{f^2 p\,\|V\|_\kappa^2}{1 - p\,\|V\|_\kappa}. \tag{10}$$

In particular, for disjoint cell-supported operators $V = \bigoplus_x V_x$, the condition is the volume-independent bound $p \sup_x \|V_x\| < 1$.

Proof. The identities $QH = I$ and $G_0 \mathcal{A}_a = I$ give

$$Q\Gamma = \Gamma Q^* = 0, \qquad \Gamma \mathcal{A}_a = I - HQ. \tag{11}$$

Consequently Γ is the inverse of the restricted positive form on $\ker Q$, extended by zero on its orthogonal complement $\mathrm{ran}\, Q^*$. The latter range is closed because Q has the bounded right inverse H.

For any curvature vector u, put $\zeta = \Gamma d^* u$ and $r = RD^*\zeta$. The constrained Euler equation is

$$\langle d\zeta - u, d\xi\rangle + \langle RD^*\zeta, RD^*\xi\rangle = 0, \qquad \xi \in \ker Q. \tag{12}$$

Set $\lambda = G'r = G'Rr$. Equation (1) gives $Q_0\lambda = 0$ and $\mathcal{L}\lambda = r$. Thus $\xi = D\lambda$ is an admissible test: $Q\xi = d_c Q_0 \lambda = 0$, $d\xi = 0$, and $RD^*\xi = Rr = r$. Equation (12) yields $\|r\|^2 = 0$. It then says that $d\zeta \in \mathcal{K}$ and $u - d\zeta \perp \mathcal{K}$, proving $d\Gamma d^* u = P_\mathcal{K} u$. At each fixed refinement all difference operators are bounded, so the same test is valid in infinite volume. No refinement-uniform norm for an individual fine derivative is needed.

Equation (5) now follows by expanding Γ. Its locality estimate follows from the two-derivative inverse bound, the locality of F, and $q_{\mathrm{aux}}^{-1} = QG_0Q^*$. In particular, no inverse of the massless precision q_M is taken. Submultiplicativity gives (6).

Flat minimality implies $P_{\mathcal{K}}F = 0$. For a real selfadjoint V with $\|V\| < 1$, every feasible curvature lies in the affine space $FB + d\ker Q$, whose closure is $FB + \mathcal{K}$. Passing to this closure does not change the infimum of the bounded strictly positive form $I + V$. The minimizing correction $k \in \mathcal{K}$ satisfies

$$(I_{\mathcal{K}} + P_{\mathcal{K}}VP_{\mathcal{K}})k = -P_{\mathcal{K}}VFB.$$

Substitution gives

$$q(V) = q_M + F^*VF - F^*VP_{\mathcal{K}}(I_{\mathcal{K}} + P_{\mathcal{K}}VP_{\mathcal{K}})^{-1}P_{\mathcal{K}}VF. \tag{13}$$

Expanding the inverse and using $P_{\mathcal{K}}^2 = P_{\mathcal{K}}$ identifies this with (7). Both V factors in (13) remain V in the complex continuation; replacing one by V^* would destroy holomorphy. The Neumann series converges in the weighted algebra for $p\,\|V\|_\kappa < 1$ and proves (9)–(10). This condition implies $\|V\| < 1$, since $p \geq 1$, on the real selfadjoint slice. Finally a cell-diagonal direct sum has $\|V\|_\kappa = \sup_x \|V_x\|$. □

Corollary 1.2 (All-order anchored curvature responses)**.** *For $n \geq 1$,*

$$D^n q(0)[V_1, \ldots, V_n] = (-1)^{n-1} \sum_{\sigma \in S_n} F^*V_{\sigma(1)}P_{\mathcal{K}}V_{\sigma(2)} \cdots P_{\mathcal{K}}V_{\sigma(n)}F, \tag{14}$$

with $n-1$ projection factors. Hence

$$\|D^n q(0)[V_1, \ldots, V_n]\|_\kappa \leq n! f^2 p^{n-1} \prod_{j=1}^{n} \|V_j\|_\kappa . \tag{15}$$

If V_j is supported wholly in curvature cell x_j, then the block from coarse cell b to coarse cell c obeys

$$\begin{aligned} &\|D^n q(0)[V_1, \ldots, V_n](b, c)\| \\ &\quad \leq f^2 p^{n-1} \prod_{j=1}^{n} \|V_j\| \sum_{\sigma \in S_n} \exp\left[-\kappa \left(\operatorname{dist}(b, x_{\sigma(1)}) + \sum_{i=1}^{n-1} \operatorname{dist}(x_{\sigma(i)}, x_{\sigma(i+1)}) + \operatorname{dist}(x_{\sigma(n)}, c)\right)\right]. \end{aligned} \tag{16}$$

Repeated source cells are allowed, and the same constants apply on compatible tori with periodic distance.

Proof. Differentiate the Neumann series in (7). Its homogeneous degree-n term has $n-1$ projections and the sign displayed in (14); polarization sums the $n!$ orders. Equation (15) follows by the algebra norm. Each block of an operator of weighted norm c is bounded by $ce^{-\kappa\,\mathrm{dist}}$. Applying this to the two F end legs and the $n-1$ projection links gives (16). The cell-supported insertions fix every intermediate cell, so there is no spatial convolution or unnormalized sum over internal fine coordinates. □

The cell-diagonal norm identity above uses disjoint cell subspaces. Overlapping supports require their actual weighted operator norm or an explicit overlap estimate. The theorem controls only the curvature metric at fixed Q; it does not supply the other terms generated by a nonflat Yang–Mills background variation.

2 Use in the construction program

A nonflat background changes more than the coefficient of the curvature quadratic form. The present theorem isolates a component whose complete response is now controlled. To apply it to an actual source family, one must identify a curvature coefficient V in the stated weighted ball and handle

the accompanying changes to the covariant derivatives, scalar gauge, corrected average and lower-order terms. A finite-regulator derivative formula alone does not verify that common domain. The auxiliary-current domain of Notebook 09 is a separate proved perturbation direction.

The conclusion concerns the local quadratic reference theory. It does not produce the interacting ultraviolet measure, restore all continuum symmetries, or establish nontriviality or a physical mass gap. Those obligations remain as recorded in Notebook 01.

References

[1] Pinned Yang–Mills-LLM-experiment account, commit `e392ef45013a581664bab76b90867170b30fe10e`.

[2] Companion research notebooks: Notebook 07, *Flat covariance locality and its free refinement limit*; Notebook 08, *A local flat minimizing extension*; and Notebook 09, *The flat fine inverse and a current neighborhood*. Their internal proofs supply the homogeneous operator estimates used here; they do not certify the complete deposited manuscript.